WHEN CAUTIOUS TEMPLATE MEMORY HURTS: A REPRODUCIBLE TRAINING-FREE STUDY OF FROZEN FARTRACKSPARSE

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Index Terms— Visual object tracking, template memory, autoregressive tracking, negative results, reproducibility.

ABSTRACT

Template updates make visual trackers adapt to changing appearance, but can also propagate a localization error. We conduct a controlled, training-free study of fourteen inference-time candidate directions for a frozen FARTrackSparse checkpoint. The development protocol uses the public GOT-10k validation set (180 sequences, 21,007 frames), while full LaSOT Testing is held out until configuration freezing. Four complete development evaluations show that hard write holding, counterfactual write rejection, stable-anchor recovery, and a logarithmic template sampler reduce mean frame IoU by 0.00823, 0.02000, 0.03204, and 0.00273, respectively, relative to the immutable baseline. Two soft rebinding variants fail fixed-screen directionality or efficiency gates. A second split-controlled phase finds that coordinate entropy is a strong failure diagnostic (AUROC 0.9067), yet seven distinct reliability- conditioned actions also fail their calibration or selection gates. Thus no proposed intervention is claimed as an improvement. The contribution of this draft is a transparent negative-result record, including checked implementation failures, statistical uncertainty, and a reproducible protocol. The frozen baseline-only LaSOT reference gives a local success AUC of 0.6148 on 280 Protocol-II sequences; no rejected candidate was run on that held-out set.

1. INTRODUCTION

Autoregressive tracking exposes a practical tension: recent templates improve adaptation, while an erroneous crop may contaminate later predictions. FARTrack uses autoregression and inter-frame sparsification to obtain fast tracking [1]. We asked a deliberately narrow question: can untrained control of its inference-time template memory improve a released, frozen FARTrackSparse checkpoint without additional training?

This paper reports the answer for the mechanisms tested: no. We regard this as useful evidence because plausible memory protections can silently damage the update distribution expected by a frozen tracker. We do not replace this negative result with a post-hoc positive claim.

2. PROTOCOL

Model and invariants. Every experiment uses the released FARTrackSparse epoch-15 checkpoint, the original prediction decoder, the original result writer, and the same initialization boxes. Its immutable baseline entry point is documented with the artifacts; the upstream inference implementation is not modified.

Development/test separation. B_{dev} is the public GOT-10k validation split. GOT-10k was introduced as a high-diversity generic tracking benchmark with an explicit one-shot protocol [2]. The development metric is mean frame IoU (AO) over 180 sequences and 21,007 frames, calculated by a read-only local script which checks sequence coverage and prediction lengths. Full official LaSOT Testing is B_{test}. LaSOT supplies dense, long sequences for SOT evaluation [3]. It was checksum-verified and not accessed until all candidate decisions were frozen. The B_{test} run is baseline-only because no candidate survived B_{dev}. It measured success AUC 0.6148, precision at 20 pixels 0.6446, and normalized precision AUC 0.6399. This is an external baseline reference, not a new-method result; no cross-run comparison to upstream repository tables is claimed.

Statistics and speed. We report the mean of per-frame IoUs. For a candidate-baseline comparison, we resample the 180 per-sequence AO differences 10,000 times with seed 20260819 and report a percentile 95% interval. Aggregate FPS is total frames divided by the sum of saved per-frame times. These intervals describe the observed development set; they are not independent-video population inference.

3. INTERVENTIONS

1. **TRM-FAR**: a transaction ledger holds writes under three causal anomaly signals and rolls back after sustained failure.
2. **Counterfactual agreement**: old-only and recent-only template views gate a write when their coordinate predictions disagree. It requires two extra forwards after warm-up.
3. **Stable-anchor recovery**: persistent failures temporarily replay the initial template and the last accepted view.
4. **Recent-consensus rebinding**: all frames are written, but fixed template slots are rebound using untrained RGB descriptor medoids.
5. **Appearance-diverse rebinding**: all frames are written, while slots retain chronological max–min RGB descriptor exemplars.
6. **Logarithmic sampler**: the upstream sampler is selected without adding a model forward or changing the write/mask implementation.

All implementations are separate tracker entry points. Focused unit tests and GPU smoke tests preceded evaluation. Counterfactual agreement initially exposed an out-of-range indexing defect after a rejected write; it was repaired by indexing its bounded template pool by accepted writes, and only the repaired version was evaluated.

Reliability follow-up. A passive probe measured coordinate-bin entropy, branch disagreement, and IoU without affecting outputs. Entropy identified frames with IoU below 0.2 with AUROC 0.9067, but a signal that predicts failure need not prescribe its repair. We therefore pre-registered a calibration/selection/confirmation partition by sequence identifier. The following actions were stopped before B_{test}: dual-lane memory, causal state mixing, delayed search expansion, rare two-view arbitration, entropy-age recent-window reading, and posterior-shape adaptive point projection. Their broad motivation overlaps uncertainty-aware tracking literature [4, 5, 6]; this study does not claim conceptual novelty for uncertainty gating.

4. RESULTS

Table 1. Complete B_{dev} results. AO is mean frame IoU.

Method	AO	Δ AO	95% paired interval	FPS
Immutable FARTrackSparse	0.832741	0.000000	–	40.07
TRM-FAR	0.822471	-0.008228	[-0.019499, 0.004104]	42.16
Counterfactual agreement	0.806470	-0.019995	[-0.037073, -0.003256]	23.47
Stable-anchor recovery	0.797473	-0.032038	[-0.050311, -0.015034]	40.90
Logarithmic sampler	0.830650	-0.002730	[-0.010716, 0.003984]	31.32

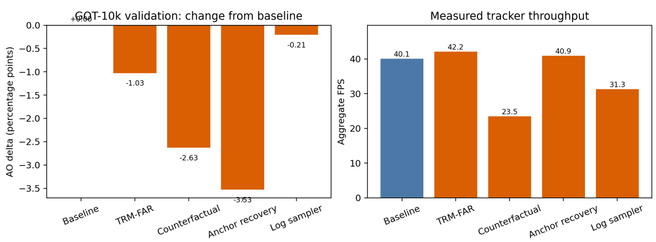

Fig. 1. Measured complete GOT-10k validation outcomes. Orange bars are rejected candidate mechanisms; the graphic reports observed B_{dev} data only and is not a held-out benchmark result.

TRM-FAR improved 70 sequences and degraded 110; stable-anchor recovery improved 66 and degraded 114; counterfactual agreement improved 77 and degraded 103. No candidate in Table 1 was tuned after its complete development result or run on B_{test}.

The two soft alternatives did not pass their prespecified focused-screen gates. Recent-consensus rebinding obtained 0.907830 versus 0.909803 for the baseline on a fixed 60-frame screen. Appearance-diverse rebinding obtained a small 0.000838 AO increase on a fixed 80-frame screen, but incurred a 16.57% FPS loss (21.93 versus 26.28 FPS), exceeding the 5% gate. These are screening observations, not generalization estimates, and were not expanded to a full development evaluation.

Table 2. Split-controlled reliability follow-up. Baseline calibration AO is 0.847678. All candidates were rejected before held-out testing.

Action	Δ AO	FPS change	Stop point
Dual-lane memory	-0.003999	-15.6%	selection
State mixing	+0.000644	-58.6%	selection
Delayed search	-0.003339	+18.3%	selection
Two-view arbitration	-0.000046	-11.6%	calibration
Recent-window read	-0.000685	+10.8%	calibration
Posterior projection	-0.009987	–	calibration

5. DISCUSSION AND LIMITATIONS

The shared pattern is that hard intervention can sacrifice normal appearance adaptation more often than it prevents memory pollution. Uncertainty, motion inconsistency, and disagreement are not calibrated error probabilities for a frozen model. Further, changing the effective template distribution may create an input regime unseen during training. For the closest-to-neutral logarithmic sampler, sequence-level AO changes have weak Spearman associations with public GOT-10k labels and simple geometric summaries (absolute $\rho \leq 0.102$ for absence, cover-label mean, cut fraction, relative motion, scale variation, and length). Thus no single observed attribute identifies when the fixed rule helps. This study does not establish that template control is never useful; it only rejects these implementations and fixed settings on the stated protocol.

Limitations are substantial. There is one released checkpoint, one full public development split, RGB descriptors rather than learned confidence, and no positive method. The LaSOT outcome is a baseline-only reference; it cannot validate a rejected candidate. GOT-10k test server submission has not been performed.

6. REPRODUCIBILITY AND DISCLOSURE

All source changes, commands, structured metrics, logs, and immutable checkpoint path are documented in the accompanying experiment log. Data archives and outputs reside on the data disk. This draft was prepared with AI assistance; human authors must verify all text, select authorship, and approve any submission. No funding, conflict-of-interest, or authorship statements are asserted because they were not provided.

7. REFERENCES

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